

Cardiovascular Health Disease Project

Dataset Link:

<https://www.kaggle.com/datasets/akshatshaw7/cardiovascular-disease-dataset>

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Idea Description:

Cardiovascular diseases are one of the leading causes of death worldwide. In this project, we aim to build machine learning models to predict whether a person has cardiovascular disease based on medical and lifestyle features. The dataset contains information such as age, blood pressure, cholesterol, weight, and physical activity.

Dataset Description:

The dataset used contains more than 30,000 records with multiple features and contains health and lifestyle information for adults such as age, blood pressure, cholesterol, weight, smoking, alcohol use, and physical activity. The target variable cardio indicates whether a person has cardiovascular disease (0 = no, 1 = yes). Machine learning classification models will be used to predict cardiovascular disease.

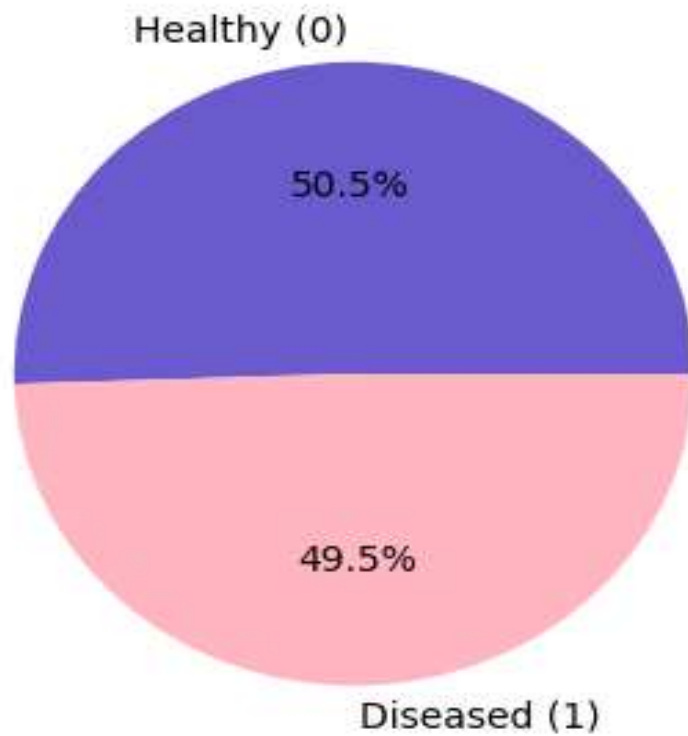
Data Preprocessing:

- Check of balanced
- Handled missing values (none found)
- Removed unnecessary columns (id, Unnamed: 0)
- Filtered unrealistic values using medical logic
- Converted age from days to years
- Handled outliers using IQR method with clipping
- Check and removed duplicate rows
- Created a new feature (BMI)

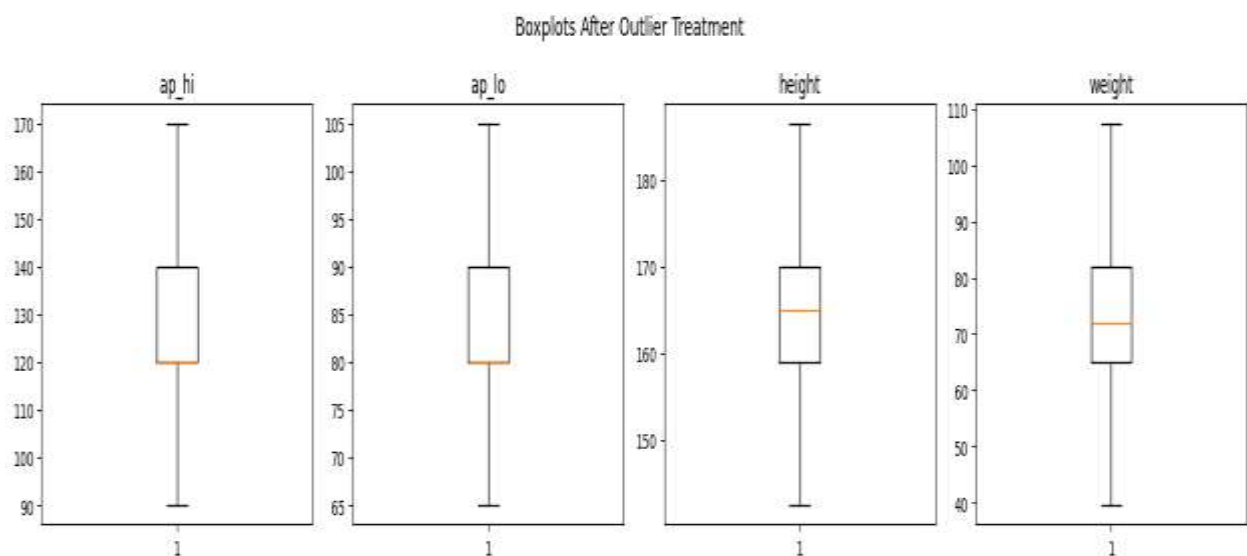
Exploratory Data Analysis (EDA)

- Visualized target distribution using pie chart

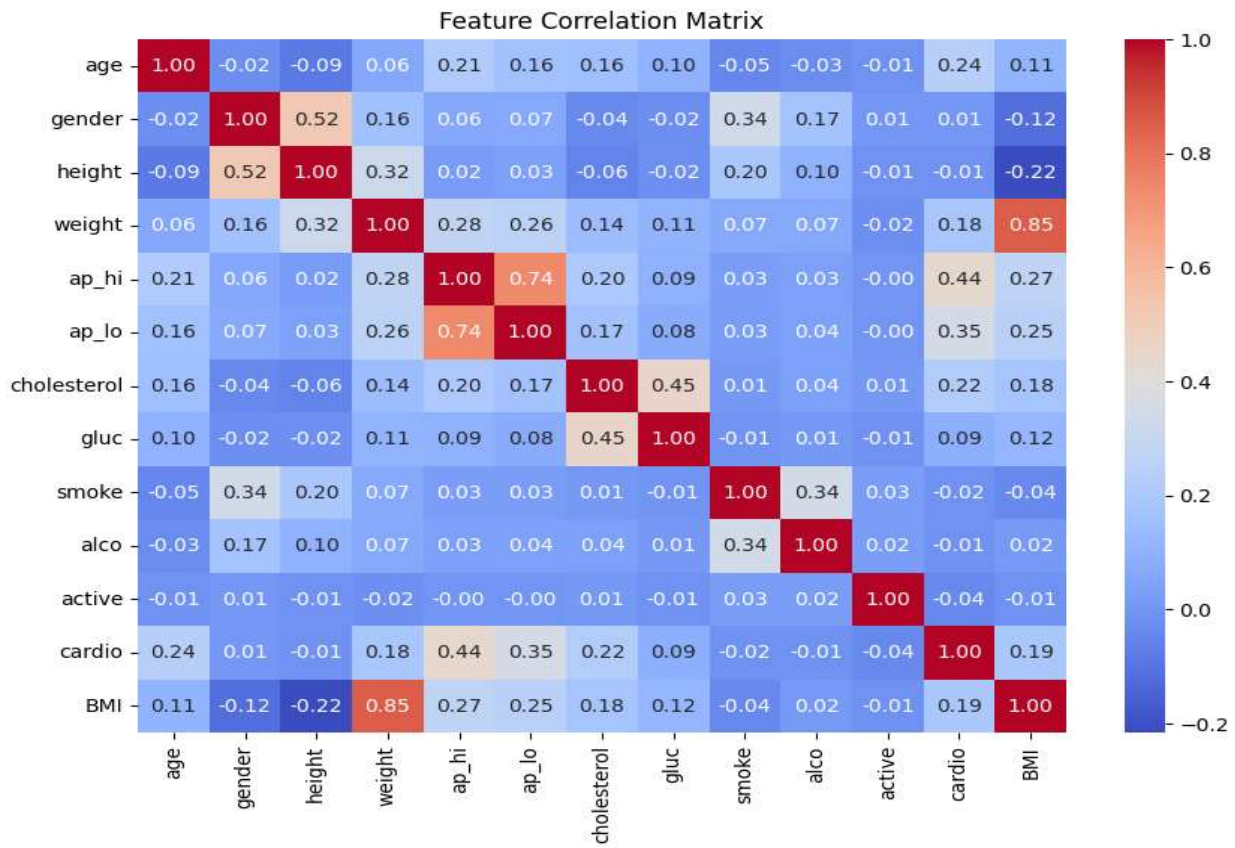
Distribution of Heart Disease Cases in Dataset



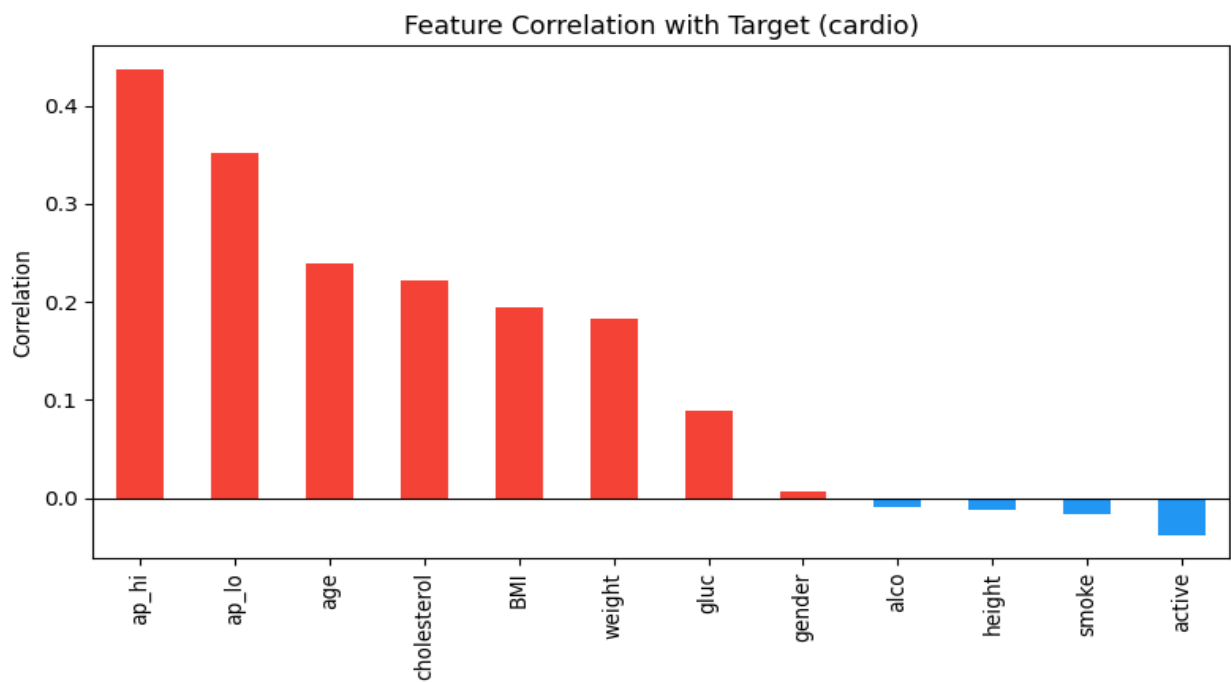
- Used boxplots to verify outlier handling



- Generated correlation heatmap



- Analyzed feature correlation with the target variable



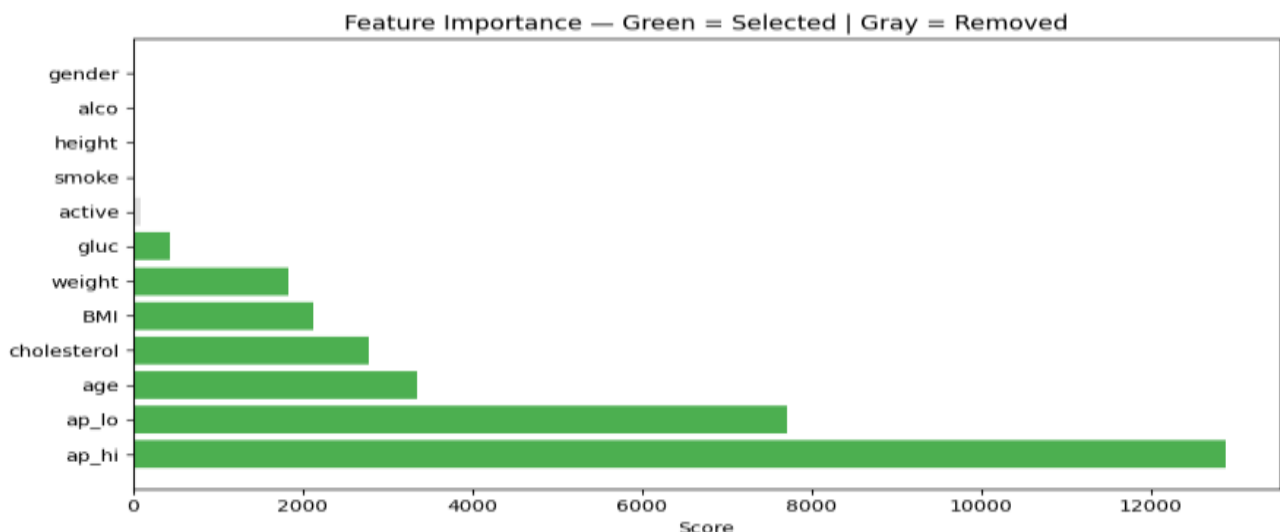
Data Splitting & Scaling

- Data was split into training (80%) and testing (20%) sets
- Stratified sampling was used
- Features were scaled using StandardScaler

Dimension Reduction

- Feature Selection

Feature Selection was applied using SelectKBest to select the most important features based on statistical significance.



```
selector = SelectKBest(score_func=f_classif, k=7)

X_train_selected = selector.fit_transform(X_train_scaled, y_train)
# Learn feature importance from training data and reduce features

X_test_selected = selector.transform(X_test_scaled)
# Apply the same feature selection to test data (no re-learning)

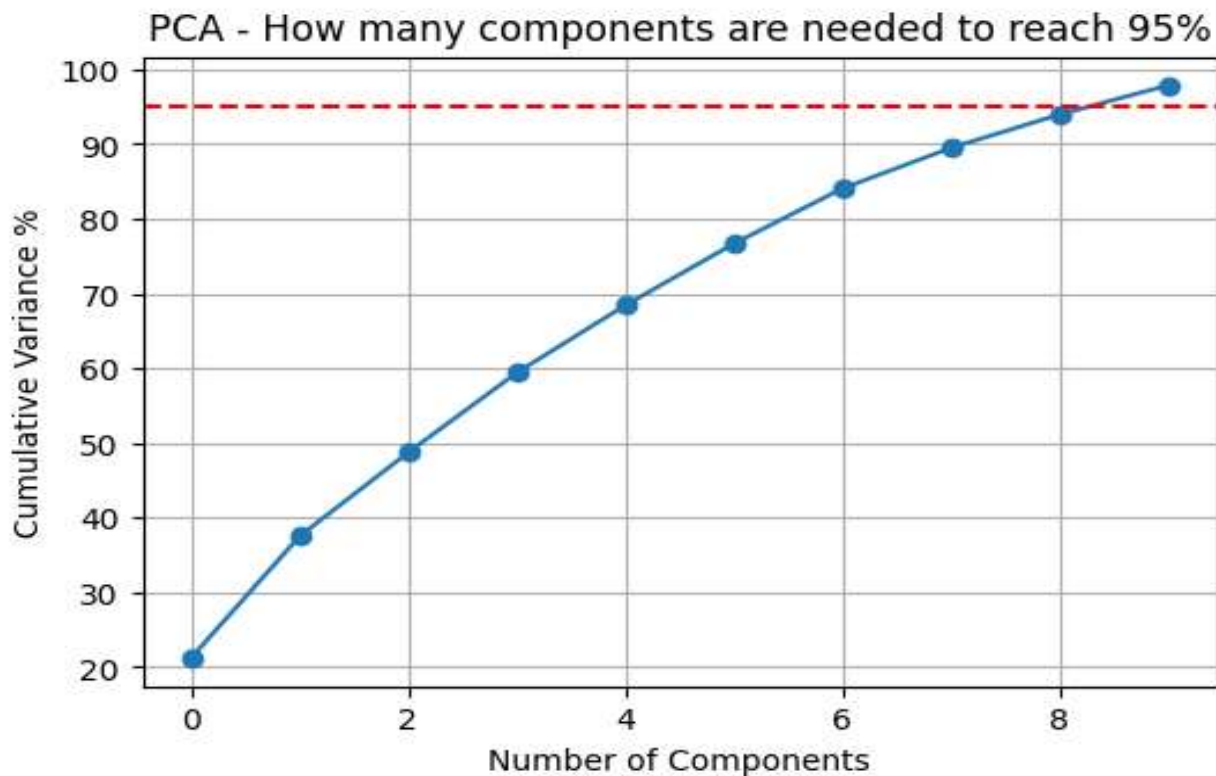
# Create a DataFrame showing feature importance ranking
scores_df = pd.DataFrame({
    'Feature': X.columns,
    'Score' : selector.scores_,
    'Selected': selector.get_support()
}).sort_values('Score', ascending=False)

print('\nFeature ranking from most important to least important:\n')
print(scores_df.to_string(index=False))

print(f'\nNumber of Features: {X_train_scaled.shape[1]} → {X_train_selected.shape[1]}')
```

- PCA (Feature Extraction)

PCA was applied to reduce dimensionality while preserving 95% of the data variance.



```
pca = PCA(n_components=0.95, random_state=42)

X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)

print(f'Number of original features      : {X_train_scaled.shape[1]}')
print(f'Number of components after PCA    : {X_train_pca.shape[1]}')
print(f'Explained information retained       : {pca.explained_variance_ratio_.sum()*100:.1f}%')
```

Modeling:

- We used three machine learning models:

Decision Tree

Random Forest

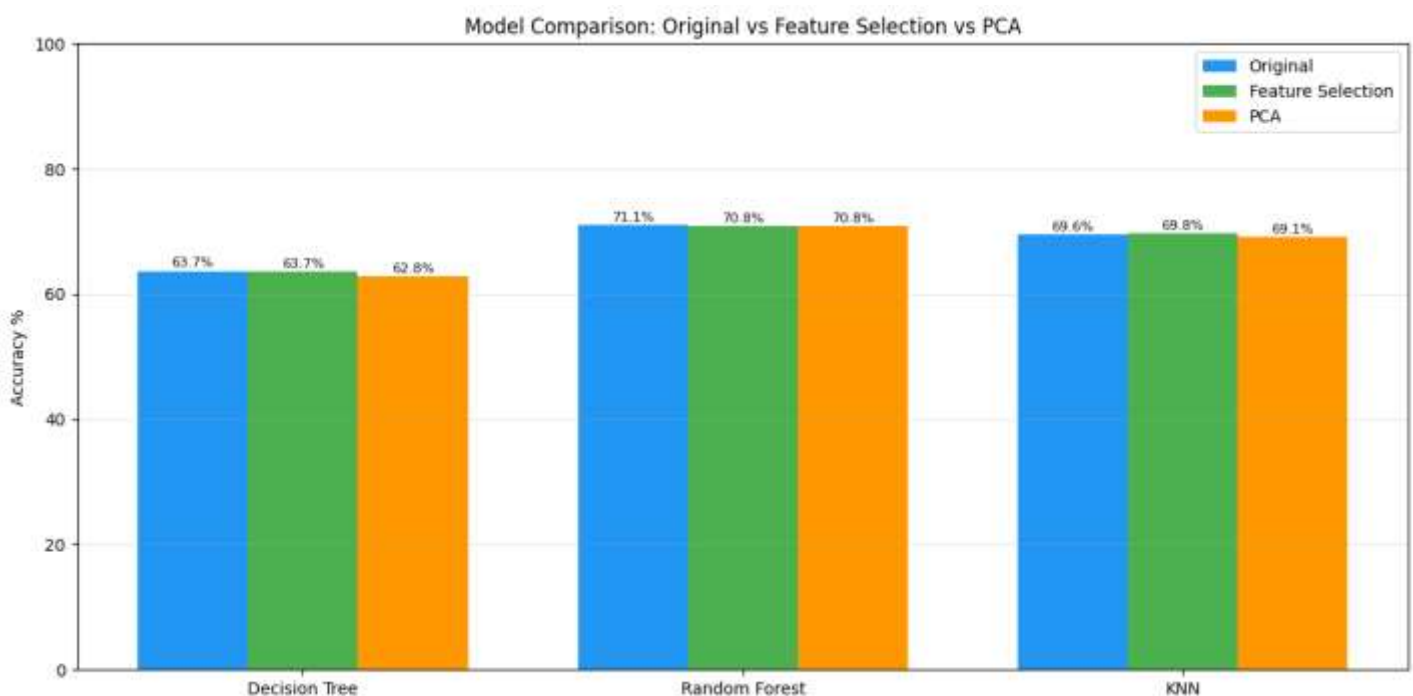
K-Nearest Neighbors (KNN)

Results & Comparison:

Models were evaluated on:

- Original Data
- Feature Selection
- PCA

The comparison showed differences in performance across different techniques.



Conclusion:

In this project, we applied machine learning techniques to predict cardiovascular disease. Feature Selection slightly improved model performance, while PCA significantly reduced dimensionality with minimal loss in accuracy. Random Forest achieved the best performance among all models.